CHAPTER – VI

PROJECT DEVELOPMENT

6.1 – Pre-Requisites

Technical Pre-Requisites

Data Architecture

- Time-Series Database: Influx DB or Timescale DB for high-frequency meter logs.
- Storage Infrastructure: AWS S3 or Google Cloud Storage for historical archives.
- Compute Engine: Apache Spark or Databricks for processing heavy data loads.

Software & Libraries

- Runtime Environment: Python 3.10+ or R 4.2+ core installations.
- Data Manipulation: Pandas, NumPy, and SciPy frameworks.
- Machine Learning: Scikit-learn, Prophet, or Boost for trend forecasting.

Data Stream Pre-Requisites

Required Metrics

- Smart Meter Telemetry: Granular load consumption recorded at 15-minute intervals.
- Environmental Factors: Local temperature, humidity, and cloud cover data.
- Socio-Demographic Layers: Building square footage, occupancy rates, and appliance types.
- Economic Indicators: Real-time electricity pricing tiers and regional peak tariffs.

Quality Standards

- Data Cleanliness: Protocols for handling missing intervals and sensor drops.
- Anomaly Rules: Filters to isolate genuine consumption surges from hardware faults.
- Timestamp Sync: Strict UTC normalization across different regional grids.

Knowledge & Skill Pre-Requisites

Domain Expertise

- Grid Mechanics: Basic understanding of peak demand, baseload, and grid strain.
- Statistical Models: Mastery of seasonal decomposition and time-series forecasting.